Contents of the shipping box

Nortek order no: 41472-1-772

Type of system: SIGNATURE 100

NORTEK
Vangkroken 2
N-1351 RUD
Norway
Tel: +47 6717 4500
Fax: +47 6713 6770
inquiry@nortek-as.com
www.nortek.no

Instrument type:

- [ ] Aquadopp
- [ ] Aquadopp 6000m
- [ ] Vector
- [ ] Signature55
- [ ] Signature1000
- [ ] Aquadopp profiler
- [ ] AWAC
- [ ] Vectrino
- [ ] Signature250
- [ ] NortekDVL
- [ ] Aquadopp 3000m
- [ ] VM AWAC
- [ ] Vectrino profiler
- [ ] Signature500

Software version: SIGNATURE DEPLOYMENT **Firmware version:** 14.4704.2206_14/165 **Other:**

Cable:

- [x] 10m length
- [ ] Other:

Connector:

- [ ] 8-pin Inline
- [ ] 12-pin UW
- [x] 6-pin Inline
- [ ] 7-pin Souriau
- [ ] Other:

Communication:

- [ ] RS232
- [x] Ethernet
- [ ] RS422
- [ ] Other:

Options:

- [ ] Analog input
- [ ] Synch
- [ ] Analog output
- [ ] Other:

Battery cannister:

- [ ] Paradopp battery cannister
- [ ] Single battery aluminium cannister
- [ ] Double battery aluminium cannister

Battery cables:

- [ ] 2pin Inline-2pin
- [ ] 8pin Inline-2pin
- [ ] 8pin rectangular-2pin

Accessories:

- [x] Toolkit
- [x] Quick guide
- [ ] Warranty card
- [x] Final test checklist
- [ ] Seeding material
- [ ] USB to serial converter RS232
- [ ] Altronix AL310 USB driver
- [ ] Recorder kit/ProLog
- [ ] Battery harness for 2 batteries

Batteries:

			Extra set
Alkaline 50Wh	13.5V	[ ]	[ ]
Alkaline 100Wh	13.5V	[ ]	[ ]
Alkaline 540Wh	13.5V	[ ]	[ ]
Alkaline 540Wh	18V	[ ]	[ ]
Alkaline 90Wh	15V(Signature1000)	[ ]	[ ]
Alkaline 180Wh	18V(Signature500)	[ ]	[ ]

AC/DC Power supply

- [ ] 15V standard
- [ ] 48V Signature55 Online
- [ ] 24V Vectrino
- [x] 24V DC/DC & Signature

Plug:

- [ ] EU
- [ ] UK
- [x] US

Other:

Date: 07.09.2018. **Responsible:** Ronny Johansen

NQM 03-080-10

Final test checklist AD2CP

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Order number: 41472-1-773

Name: Signature 100

Instrument serial number: 101132

Frequency: 100KHz

Main board: AD2CP_1252

Firmware versions: 1.4.4704.2206_14/165

Label checked [X] OK
Dock test [X] OK
Baudrate 115200 [X] OK

Comments:

Tilt check
[X] Pitch up
[X] Roll up
[X] Status bit
[X] Pitch down
[X] Roll down

pitch & roll within +/- 0.2 °

Clock
[X] Set clock

Heading
[X] Up
[X] Down

tolerance: +/- 2 °

Pressure
1
100
101

Psensortemp
[X] OK

tolerance: +/- 0.1 % of 1500 m

Temperature
[X] OK

tolerance: +/- 0.1 °

Beam check

	Correct order	Beam Imp	Noise floor	Amplitude in tank	Range
Beam 1	[X] OK	84 Ω	25 dB	> 80 dB	[X] OK
Beam 2	[X] OK	71 Ω	25 dB	> 80 dB	[X] OK
Beam 3	[X] OK	68 Ω	25 dB	> 80 dB	[X] OK
Beam 4	[X] OK	69 Ω	25 dB	> 80 dB	[X] OK
Beam 5	[X] OK	568 Ω	25 dB	> 80 dB	[X] OK

Velocity direction
To heavy to Verify

Matrix Correct.

XYZ coordinate system		ENU coordinate system	
X	[ ] OK	E	[ ] OK
Y	[ ] OK	N	[ ] OK
Z	[ ] OK	U	[ ] OK
Z_2	[ ] OK	U_2	[ ] OK

Head file
[X] Headfile checked
[X] Saved as read only

Serial communication
~~[ ] RS422~~
~~[ ] RS232~~

Recorder erased
[X] OK

Rec size: 128GB

Ethernet
MAC address: 8C:68:78:00:04:E4

Static IP address:

Set host name: 101132

DHCP enabled [X]

FTP OK [X]

Licenses

Averaging mode	[X]	Wave mode	[ ]	Vertical velocity	[ ]	64GB recorder	[ ]
Burst Five beams	[ ]	Echo Sounder	[X]	Dual frequency low	[ ]	128GB recorder	[X]
Bottom track	[ ]	Ice Measurement	[ ]	Dual frequency high	[ ]	256GB recorder	[ ]
High Resolution	[ ]	Altimeter	[ ]	16GB recorder	[ ]		

Calibration license erased [X]
Production license erased [X]
Default configuration set [X]

Cable/Harness

	Cable		Harness
Communication	[X]	Communication	[X]
Battery	[ ]	Battery	[ ]

Electrical isolation test
50V Ok [X]

External sensors

Power down
[X] OK

Date
Day 7 Month 9 Year 18

Signature

Certificate of Calibrations and Tests

Instrument Information

Customer Reference No.	41472-1-773
Instrument Type	Signature100
Instrument Frequency	100 kHz
Instrument S/N	101132
Head S/N	D-1132
Interface Board S/N	1252
Interface Board Mfr. S/N	4MO0476290007
Digital Board Mfr. S/N	4MO0344030020
Analog Board Mfr. S/N	
Analog Board #2 Mfr. S/N	
Sensor Board Mfr. S/N	4MO0319780037
Interface Board Rev.	H-1(High-Power)
Digital Board Rev.	I-3
Analog Board Rev.	B-0(High-Power Echo)
Analog Board #2 Rev.	
Sensor Board Rev.	I-0

Calibrations and tests performed

Pressure	Passed
Tilt and Compass	Passed

All the tested values are within Nortek AS specifications

September 3, 2018
Date

Reviewed and approved (sign.)

Details

Instrument Type	Signature100
Instrument S/N	101132
Pressure Range	1500 dBar
Date	August 31, 2018
Operator	Daniel Røyert
Location	Nortek Factory Norway
Result	Passed

Description

Verification is performed in an automated pressure chamber. Fixed-point measurements are collected to verify the sensor.

Criteria of acceptance is ± 0.1% of full scale.

Reference: Telemark Technologies - Vessel Pressure Sensor Inlet.

Verification Results

Reference (dBar)	Pressure Diff. (dBar)	Pressure Diff. (% of FS)
331.67	-1.32	-0.09
627.63	-0.87	-0.06
922.02	-0.82	-0.05
1222.91	0.46	0.03
1517.20	1.32	0.09

Tilt and Compass Report

Details

Instrument Type	Signature100
Instrument S/N	101132
Date	August 28, 2018
Operator	Asle Martinsen
Location	Nortek Factory Norway
Result	Passed

Description

Calibration and verification is performed in a two axis automated jig. Continuous and fixed-point measurements are collected to calibrate and verify the sensor.

Criteria of acceptance for tilt sensor is ±0.2°.

Criteria of acceptance for compass sensor is ±2°.

Reference: Digital Protractor Series 950 Pro 3600. Accuracy ±0.05°.

Tilt Verification Results

	Diff. Up		Diff. Down	
Reference (°)	**Pitch (°)**	**Roll (°)**	**Pitch (°)**	**Roll (°)**
-30.00	-0.16	-0.07	-0.16	0.11
-15.00	0.00	-0.04	0.03	-0.08
0.00	0.17	-0.03	0.17	-0.07
15.00	-0.17	0.06	-0.17	0.11
30.00	-0.04	-0.10	0.03	0.00

Compass Verification Results

Reference (°)	Heading Diff. Up (°)	Heading Diff. Down (°)
0.00	0.36	0.48
45.00	0.07	-0.23
90.00	-0.09	-0.58
135.00	-0.06	-0.33
180.00	0.05	0.28
225.00	0.29	1.19
270.00	0.59	1.34
315.00	0.24	1.16